

Chaithanya Virupaksha

Forward Deployed Engineer · Python & Client Delivery

Right to work UK Graduate Route from Dec 2026 · no sponsorship needed to start

Newcastle upon Tyne · open to relocating anywhere in the UK chaithanya6699@gmail.com
+44 7799 042262 github.com/chai6699 chai6699.github.io [LinkedIn](#)

PROFILE

MSc Advanced Computer Science graduate (Newcastle University) who enjoys working where customers and code meet. For 15 months full-time I worked directly with clients: scoping what they needed, adapting machine-learning models to their data and explaining the results to non-technical users, then handing each project over. On the engineering side, I built the React front end and browser-side Keycloak authentication for a Spring Boot microservices platform on Azure Kubernetes Service, and a Python research framework covered by 50 unit tests. I like owning a problem from the first conversation to a working handover.

2.5 yrs industry ML, incl. 15 months full-time delivering client models	98% held-out accuracy, OpenCV object- detection pipeline	50 unit tests on a reproducible FL simulator	~25% Lighthouse performance gain from a front-end rebuild
---	--	--	---

EXPERIENCE

Machine Learning Engineer May 2024 – Aug 2025

Accurate Info Solution · Bengaluru

- Worked directly with clients, mostly engineering colleges, to scope requirements, adapt models to their data and explain results to non-technical users.
- Took each client project from requirements through data preparation, model training and evaluation to handover.
- Progressed from intern to engineer, returning after graduation to own client ML projects end to end.

Full-Stack Development Intern Sep 2023 – Mar 2024

PACE Wisdom Solutions · Bengaluru

- Shipped responsive features in two-week Agile sprints alongside design and back-end engineers.

SKILLS

CORE FOR THIS ROLE

- Python
- JavaScript
- React 19
- REST
- OpenAPI/Swagger
- SQL
- Docker
- Kubernetes
- GitHub Actions CI/CD
- Keycloak
- OAuth2/JWT

WORKING WITH CUSTOMERS

- Requirements scoping
- Stakeholder communication
- Technical handover
- Agile delivery

LANGUAGES

- Java 21
- C++
- C

ML & DATA

- PyTorch
- scikit-learn
- NumPy
- Pandas
- OpenCV
- NLP (TF-IDF, LDA)
- Model evaluation

SOFTWARE

- Spring Boot 3
- Quarkus
- Flask
- HTML/SCSS
- JPA/Hibernate
- PostgreSQL
- MySQL
- Unit testing
- Microservices

CLOUD & DEVOPS

- Microsoft Azure
- Azure Kubernetes Service
- NGINX Ingress
- Prometheus
- Grafana
- AWS S3
- Git

SECURITY

- RBAC
- Secure SDLC

- Rebuilt front-end components in HTML, SCSS and Bootstrap on a component-based architecture, raising Lighthouse performance scores by around 25%.

Machine Learning Intern (part-time, alongside BEng) Sep 2022 – Dec 2023

Accurate Info Solution · Bengaluru

- Built a Python and OpenCV object-detection pipeline across multiple object categories, reaching 98% classification accuracy on a held-out test set.
- Prototyped a CNN for plant-disease identification from leaf images that outperformed the rule-based baseline it replaced.

PROJECTS

Cloud-Native Bookstore Platform 2025–2026

Team of six · Newcastle University

- Built the React front end (catalogue, book detail, cart and checkout) for a microservices platform of Catalog, Cart and Gateway services on Azure Kubernetes Service.
- Implemented browser-side authentication with Keycloak: parsed JWT claims into application state, refreshed tokens silently before expiry, and gated routes and admin actions by role.
- Made cart interactions feel instant with optimistic updates that refetch from the API on failure.
- Co-authored the architecture report covering blue-green releases, autoscaling and Velero-based recovery.

React 19

Keycloak

Spring Boot 3

Azure Kubernetes Service

GitHub Actions

AI Bias Audit Framework for Generative AI 2025–2026

Newcastle University

- Built a framework that measures gender, cultural and role-stereotype bias across several LLMs using TF-IDF, LDA topic modelling and Shannon-entropy diversity scores, with a fixed prompt set so each run is a repeatable measurement.

Python

LLM evaluation

scikit-learn

NLP

Federated Learning under Lossy Wireless Networks 2026

MSc dissertation · Newcastle University · [GitHub repository](#)

- Found that compression does more than save bandwidth: it decides which devices can take part. The accuracy gap between selection strategies fell from 16.4 to 2.3 percentage points once updates arrived within the round deadline.
- Audited my own work, found and fixed eight defects, and documented the ones that changed or qualified results in the repository changelog.

EDUCATION

MSc Advanced Computer Science

Newcastle University · 2025–2026

Dissertation submitted Sept 2026; award expected Dec 2026. Cloud Computing, Secure Software Development, Risk & Trust Management, Generative AI for Businesses, Enterprise Software & Services, Advanced Programming in Java.

BEng Computer Science

Dr. Ambedkar Institute of Technology, Bengaluru · 2024

CGPA 8.3 / 10

PUBLICATIONS

Virupaksha, C. (2026). Privacy-Preserving Digital Twin Models for Smart Healthcare Applications. *IJDTSC*, 2(1).

Siddaraju, Dr., Virupaksha, C. et al. (2024). Smart Irrigation Using Embedded AI. *IJIRCCE*, 12(6).

CERTIFICATIONS

Python 3 Programming, University of Michigan
Databases and SQL for Data Science with Python

BEYOND CODE

Bharatanatyam performer with Newcastle University's Indian Performing Arts Society.

Organised Kalarava 2024, a college festival, coordinating volunteers and logistics across multiple events.

LANGUAGES

English (full professional) · Kannada (native) · Hindi (limited working)